120 Days of Quantitative Finance

Month : Advanced Time Series Modelling for Finance

Week 15: Advanced Time Series Modelling

Day 100: Stationarity and Differencing

1 Introduction

Stationarity is a fundamental concept in time series analysis. Most statistical and econometric models assume that the underlying process is stationary.

2 Definition of Stationarity

A time series $\{X_t\}$ is said to be **strictly stationary** if:

$$(X_{t_1}, X_{t_2}, ..., X_{t_n}) \stackrel{d}{=} (X_{t_1+h}, X_{t_2+h}, ..., X_{t_n+h})$$

for all t and lag h.

3 Weak Stationarity

In practice, we use weak (second-order) stationarity:

- Constant mean:

$$E[X_t] = \mu$$

- Constant variance:

$$Var(X_t) = \sigma^2$$

- Covariance depends only on lag:

$$Cov(X_t, X_{t+h}) = \gamma(h)$$

4 Non-Stationary Series

A series is non-stationary if:

- mean changes over time (trend)
- variance changes (heteroskedasticity)
- structural breaks occur

Financial price series are typically non-stationary.

5 Differencing

To make a series stationary, we apply differencing:

$$\Delta X_t = X_t - X_{t-1}$$

Higher-order differencing:

$$\Delta^2 X_t = \Delta(\Delta X_t)$$

6 Log Returns

In finance, we often use log returns:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Log returns are often stationary even when prices are not.

7 Unit Root Concept

A non-stationary process often contains a unit root:

$$X_t = X_{t-1} + \epsilon_t$$

Differencing removes the unit root.

8 Why Stationarity Matters

Stationarity ensures:

- stable statistical properties
- meaningful model estimation
- reliable forecasting

9 Conclusion

Stationarity is a prerequisite for most time series models.

Differencing and transformations allow us to convert non-stationary data into a usable form.